

SHRINISH VHANBATTE

📞 7350032932 ✉ shrinishiitkgp2001@gmail.com 💻 Shrinish Vhanbatte 🌐 shrinish123

EDUCATION

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

2019 - 2024

Dual Degree (B.Tech + M.Tech) in Civil Engineering

8.16/10 CGPA

EXPERIENCE

TestMu AI (formerly LambdaTest) — Member of Technical Staff | Go, Python, C++, AWS

April 2024 – Present

- Engineered a high-performance **Visual Engine** from scratch in **C++ with OOPs** (Python, AWS Lambda, Docker) — **smart-comparison algorithms** that suppress rendering noise and false positives, plus a layout-diff renderer wired across **Go, JS, and Python** microservices over **Kafka** — boosting product revenue by **20%**.
- Engineered and developed a **robust scheduler microservice** to manage and schedule web scans using cron jobs efficiently, integrating with MySQL and Kafka for **event-driven communication**.
- Built CRUD APIs in Go microservices to support comparison techniques, resolved multiple customer issues, addressed **race conditions**, and **optimized SQL queries** to improve performance by **40%**.
- Engineered a **multi-baseline screenshot comparison** subsystem — designed the variation schema, **Redis-coordinated** parallel fan-out via **Go goroutines**, and deterministic best-match resolution wired through dashboard and public APIs.
- Added **network logs** support in **KaneAI** real-device sessions by integrating a **MITM proxy binary** into the **HyperExecute VM** flow, capturing full HTTP traffic to power AI-driven test authoring and customer debugging.

TCG Digital

May 2023 – Jul 2023

SDE Intern

Kolkata, India

- Built a web app that generates **source code and Dockerfiles** for CRUD applications from user-supplied specs using **React, Node.js, PostgreSQL**.

PROJECTS

Capia — Autonomous Agentic Commerce | Python, FastAPI, Postgres, Next.js, MCP, WebAuthn | Live | Github

August 2026

- Built an agent that buys unattended: natural language compiles into a **typed purchase rule**, a watcher polls live merchant catalogues over **Shopify UCP** (23 stores), and on a price/stock hit it quotes, charges, and buys — inside a spend cap enforced by the **Visa network** via passkey-approved mandates, not by application code.
- Designed **capability-scoped agent authority** (tools to raise a limit or approve spend simply do not exist) across four surfaces — web chat, voice, **WhatsApp**, and an **MCP server** over streamable HTTP with scoped revocable tokens — on **FastAPI + Postgres**, **286 passing tests**, one-command Docker bring-up.

AgentCI — CI/CD & Evals for AI Agents | Python, FastAPI, Kafka, ClickHouse, MCP | Github

July 2026 – Present

- Building a **regression-testing and eval pipeline for AI agents**: eval suites fan out across a model × parameter matrix, scored by a **two-stage LLM-as-judge**, with baseline diffs, **flakiness scoring**, and a pass/warn/fail **CI gate**.
- Designed the distributed backbone — **FastAPI control plane** (REST, WebSocket, **MCP server**), **Kafka** workers with DLQ, **Postgres OLTP + ClickHouse** analytics — with per-provider **rate limiting** and per-run **cost budgets**.

Fivo — Voice-First Finance Agent | Python, FastAPI, Postgres (pgvector), Redis, Celery, Docker | Github

May 2026 – Present

- Built the backend for a voice-first finance product — **async FastAPI + SQLAlchemy 2**, JWT auth, **Redis, Celery, MinIO/S3**, Dockerized — with a **tool-calling voice agent** that runs account, transaction, and budget operations from speech, backed by a **RAG pipeline** over **pgvector**.

ACHIEVEMENTS / HONORS

- Won **Gold** at **Inter-IIT Tech Meet 10.0** (Mercari large-scale system design) for a cloud-native microservices architecture for a Healthcare Management System; **Best Feature Prize** at the IIT Kharagpur Website Hackathon among 30+ teams.
- Maximum rating of **2140** (top 1.46%) on **LeetCode**, 1812 on CodeChef, 1529 on Codeforces (*handle: bhatte*).
- Global rank **865** and **1440** in **Google Kick Start** Round H / Round B 2022; rank **1531** in **Meta Hacker Cup** Round 2 (T-shirt awarded to top 2000).
- Open source**: mentored 10+ **KOSS** students (2021 & 2022) on projects like Mumble and letsUpsolve; contributed issues and PRs to **Rocket.Chat**.

SKILLS

- Languages**: Go, C++, Python, JavaScript/TypeScript, SQL
- Backend & Infra**: Microservices, REST, gRPC, Kafka, Redis, MySQL, PostgreSQL, AWS (Lambda, S3), Docker, Kubernetes
- Monitoring & Tooling**: Datadog, Sumo Logic, New Relic, Git, GitHub Actions, CI/CD
- AI / ML**: LLM agents & tool calling, RAG, LLM-as-judge evals, MCP, LangChain, OpenAI / Anthropic APIs